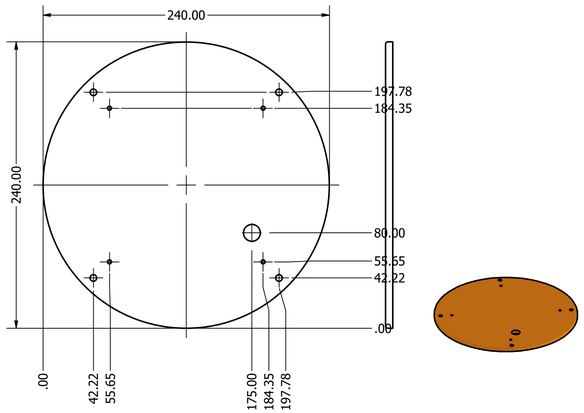
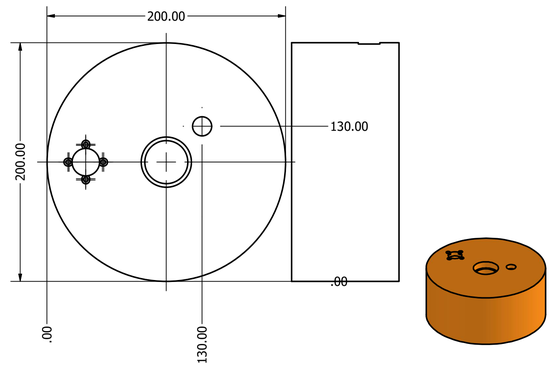
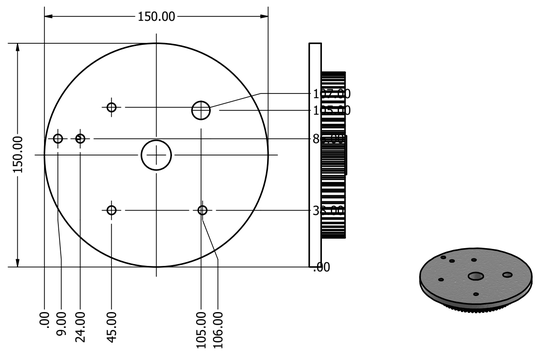
RA-101 BASEPLATE
PETG, QTY 1, 341 g (INVENTOR)
SCALE 1:2.5



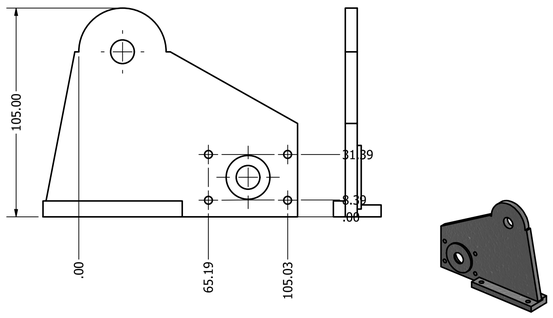

RA-102 BASEHOUSING
PETG, QTY 1, 611 g (INVENTOR)
SCALE 1:2.5



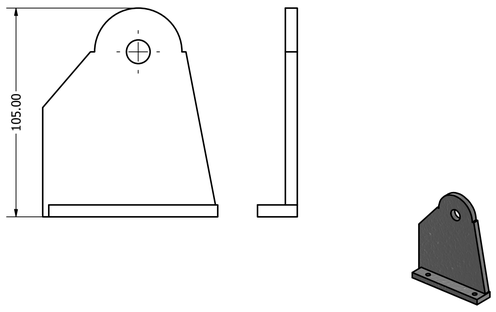

RA-103 TURNTABLE
PA6-CF, QTY 1, 237 g (INVENTOR)
SCALE 1:2




RA-104 SHOULDERCHEEKR
PA6-CF, QTY 1, 70 g (INVENTOR)
SCALE 1:1.5



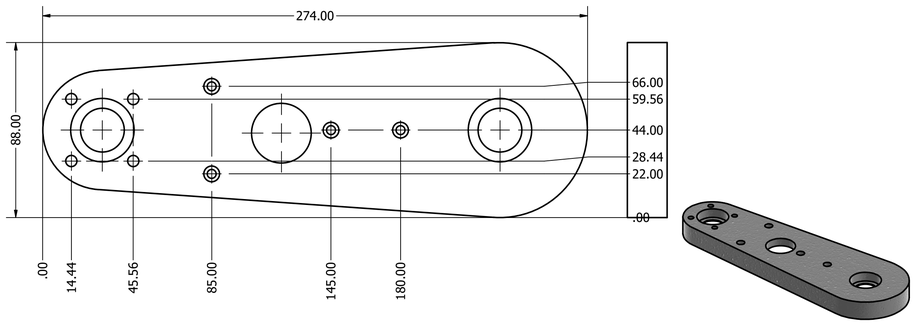

RA-105 SHOULDERCHEEKL
PA6-CF, QTY 1, 53 g (INVENTOR)
SCALE 1:1.5



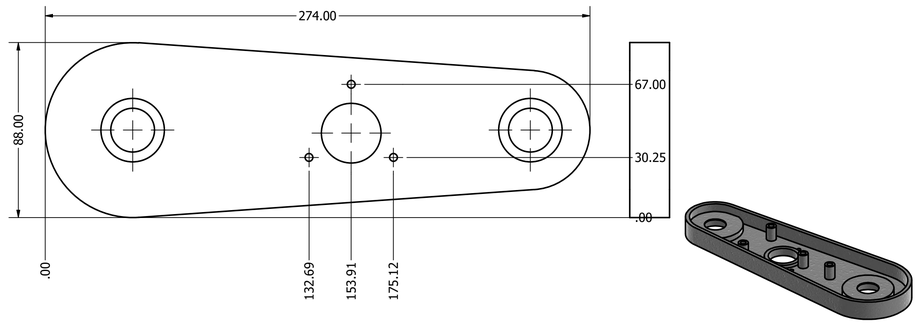

NOTES:
1. MATERIAL: PA6-CF (BAMBU LAB) UNLESS NOTED; PETG WHERE NOTED. DRY PA6-CF 80 C / 8 H.
2. PRINT: 0.4 MM NOZZLE, 0.2 MM LAYERS, 5 WALLS, 40% GYROID, 6 TOP/BOTTOM. ANNEAL PA6-CF 80 C / 6 H.
3. BEARING SEATS PRINTED -0.10 MM, PRESS FIT AFTER TEST PRINT. M3/M4 BRASS HEAT-SET INSERTS WHERE 4.0 / 5.6 HOLES.
4. GEARS: 20 DEG PRESSURE ANGLE, PRINT FLAT (TEETH IN XY). UNTOLERANCED DIMS +/-0.2 MM.
5. ALL DIMENSIONS MM. SOURCE: _build/design.py + build_parts.py (PARAMETRIC SCRIPT).

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|---|---|---|
| DRAWN Jay Patel | 11/1/2026 | |
| CHECKED | | TITLE: RoboticArm6DOF - printed part manufacturing drawings |
| QA | | |
| MFG | | |
| APPROVED | | |
| | | SIZE D · DWG NO RA-DWG-100 · REV |
| | | SCALE 0.40 : 1 · SHEET 1 OF 5 |

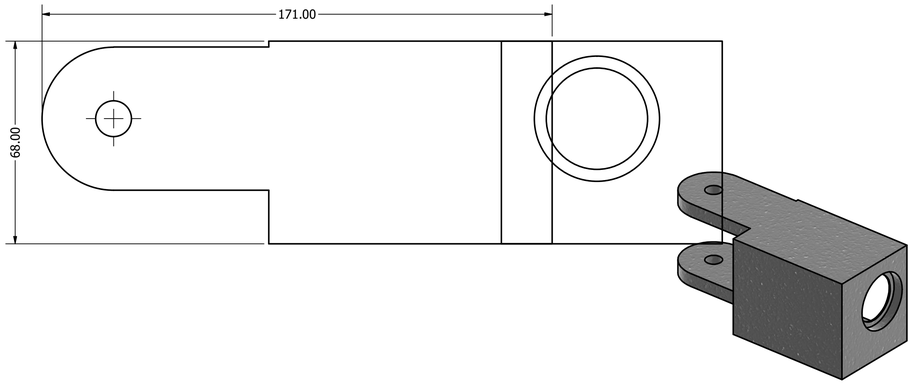
RA-106 UPPERARMA
PA6-CF, QTY 1, 156 g (INVENTOR)
SCALE 1:1.5



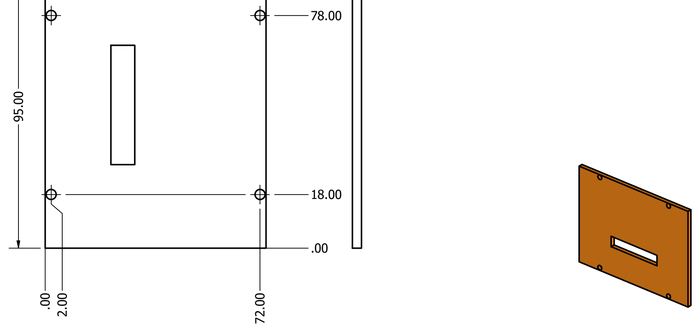

RA-107 UPPERARMB
PA6-CF, QTY 1, 157 g (INVENTOR)
SCALE 1:1.5




RA-108 FOREARM
PA6-CF, QTY 1, 202 g (INVENTOR)
SCALE 1:1



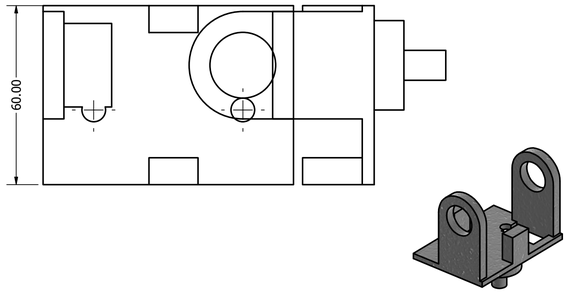

RA-109 FOREARMCOVER
PETG, QTY 1, 25 g (INVENTOR)
SCALE 1:1



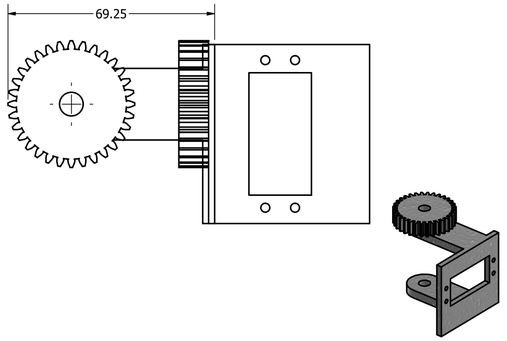

NOTES:
1. MATERIAL: PA6-CF (BAMBU LAB) UNLESS NOTED; PETG WHERE NOTED. DRY PA6-CF 80 C / 8 H.
2. PRINT: 0.4 MM NOZZLE, 0.2 MM LAYERS, 5 WALLS, 40% GYROID, 6 TOP/BOTTOM. ANNEAL PA6-CF 80 C / 6 H.
3. BEARING SEATS PRINTED -0.10 MM, PRESS FIT AFTER TEST PRINT. M3/M4 BRASS HEAT-SET INSERTS WHERE 4.0 / 5.6 HOLES.
4. GEARS: 20 DEG PRESSURE ANGLE, PRINT FLAT (TEETH IN XY). UNTOLERANCED DIMS +/-0.2 MM.
5. ALL DIMENSIONS MM. SOURCE: _build/design.py + build_parts.py (PARAMETRIC SCRIPT).

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| DRAWN Jay Patel | 11/1/2026 | |
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| QA | | TITLE: RoboticArm6DOF - printed part manufacturing drawings |
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| APPROVED | | |
| | | SIZE D · DWG NO RA-DWG-100 · REV |
| | | SCALE 2 / 3 · SHEET 2 OF 5 |

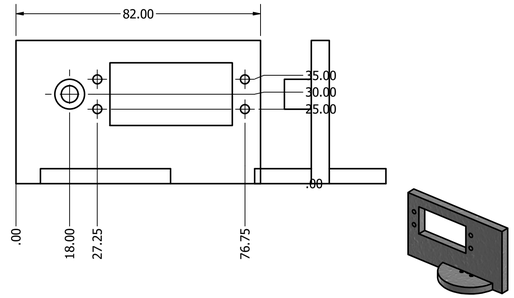
RA-110 ~~WRISTYOKE~~
PA6-CF, QTY 1, 59 g (INVENTOR)
SCALE 1:1



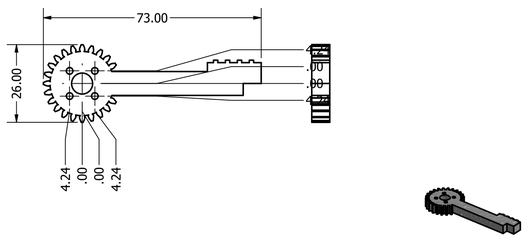

RA-111 ~~WRISTBRACKET~~
PA6-CF, QTY 1, 39 g (INVENTOR)
SCALE 1:1



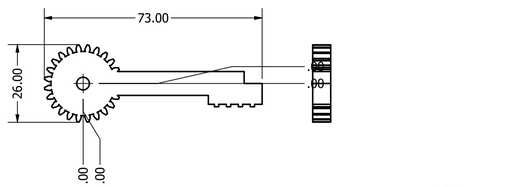

RA-112 ~~GRIPPERBASE~~
PA6-CF, QTY 1, 27 g (INVENTOR)
SCALE 1:1




RA-113 FINGERL
PA6-CF, QTY 1, 5 g (INVENTOR)
SCALE 1:1



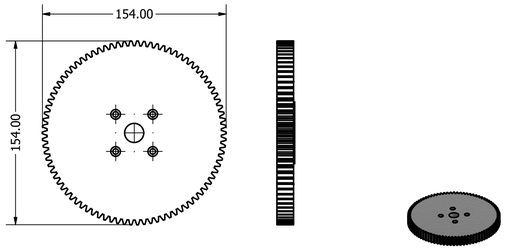

RA-114 FINGERR
PA6-CF, QTY 1, 6 g (INVENTOR)
SCALE 1:1



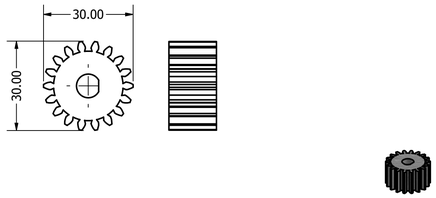

NOTES:
1. MATERIAL: PA6-CF (BAMBU LAB) UNLESS NOTED; PETG WHERE NOTED. DRY PA6-CF 80 C / 8 H.
2. PRINT: 0.4 MM NOZZLE, 0.2 MM LAYERS, 5 WALLS, 40% GYROID, 6 TOP/BOTTOM. ANNEAL PA6-CF 80 C / 6 H.
3. BEARING SEATS PRINTED -0.10 MM, PRESS FIT AFTER TEST PRINT. M3/M4 BRASS HEAT-SET INSERTS WHERE 4.0 / 5.6 HOLES.
4. GEARS: 20 DEG PRESSURE ANGLE, PRINT FLAT (TEETH IN XY). UNTOLERANCED DIMS +/-0.2 MM.
5. ALL DIMENSIONS MM. SOURCE: _build/design.py + build_parts.py (PARAMETRIC SCRIPT).

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| MFG | | |
| APPROVED | | |
| | | SIZE D · DWG NO RA-DWG-100 · REV |
| | | SCALE 1 : 1 · SHEET 3 OF 5 |

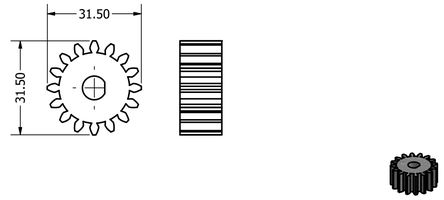
RA-203 J2GEAR86T
PA6-CF, QTY 1, 183 g (INVENTOR)
SCALE 1:2.5



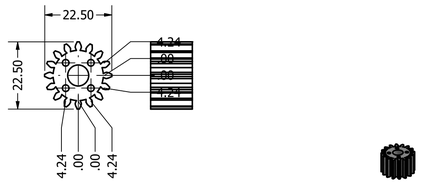

RA-204 J1PINION18T
PA6-CF, QTY 1, 9 g (INVENTOR)
SCALE 1:1



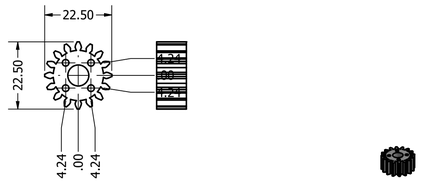

RA-205 J2PINION16T
PA6-CF, QTY 1, 8 g (INVENTOR)
SCALE 1:1



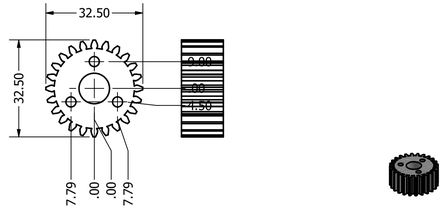

RA-206 J4PINION
PA6-CF, QTY 1, 4 g (INVENTOR)
SCALE 1:1



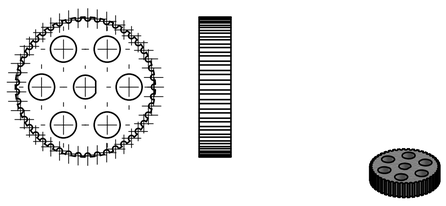

RA-207 J5PINION
PA6-CF, QTY 1, 3 g (INVENTOR)
SCALE 1:1




RA-208 J4GEAR
PA6-CF, QTY 1, 9 g (INVENTOR)
SCALE 1:1




RA-209 PULLEYHTD5M_45T
PA6-CF, QTY 1, 48 g (INVENTOR)
SCALE 1:1.5

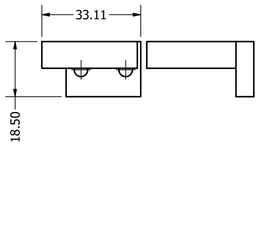


NOTES:
1. MATERIAL: PA6-CF (BAMBU LAB) UNLESS NOTED; PETG WHERE NOTED. DRY PA6-CF 80 C / 8 H.
2. PRINT: 0.4 MM NOZZLE, 0.2 MM LAYERS, 5 WALLS, 40% GYROID, 6 TOP/BOTTOM. ANNEAL PA6-CF 80 C / 6 H.
3. BEARING SEATS PRINTED -0.10 MM, PRESS FIT AFTER TEST PRINT. M3/M4 BRASS HEAT-SET INSERTS WHERE 4.0 / 5.6 HOLES.
4. GEARS: 20 DEG PRESSURE ANGLE, PRINT FLAT (TEETH IN XY). UNTOLERANCED DIMS +/-0.2 MM.
5. ALL DIMENSIONS MM. SOURCE: _build/design.py + build_parts.py (PARAMETRIC SCRIPT).

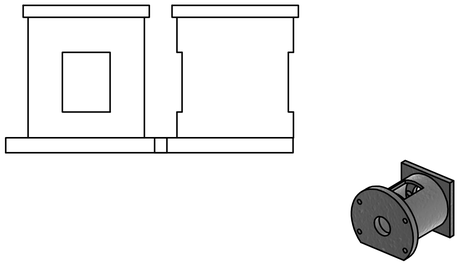
GEAR DATA (20 DEG PRESSURE ANGLE, STANDARD ADDENDUM, PRINTED PA6-CF)
J1: m1.5 18T:72T RATIO 4.000 CENTRE 67.50 FACE 16 PCD 27.00 / 108.00
J2: m1.75 16T:86T RATIO 5.375 CENTRE 89.25 FACE 14 PCD 28.00 / 150.50
J4: m1.25 16T:24T RATIO 1.500 CENTRE 25.00 FACE 14 PCD 20.00 / 30.00
J5: m1.25 16T:32T RATIO 2.000 CENTRE 30.00 FACE 10 PCD 20.00 / 40.00
GRIP: m1 24T:24T RATIO 1.000 CENTRE 24.00 FACE 6 PCD 24.00 / 24.00
J3: HTD 5M 15T:45T BELT 375-5M-15, CENTRE 109.91 (ECCENTRIC 608 HOUSINGS, +-1.5 MM TENSION)
J1 LEWIS: PINION 17.8 MPA, GEAR 12.7 MPA (ALLOW 30)
J2 LEWIS: PINION 12.3 MPA, GEAR 8.2 MPA (ALLOW 30)
J4 LEWIS: PINION 10.8 MPA, GEAR 9.5 MPA (ALLOW 30)
J5 LEWIS: PINION 11.5 MPA, GEAR 9.3 MPA (ALLOW 30)

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| DRAWN Jay Patel | 11/1/2026 | TITLE: RoboticArm6DOF - printed part manufacturing drawings |
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| MFG | | |
| APPROVED | | |
| | | SIZE D \| DWG NO RA-DWG-100 \| REV |
| | | SCALE 0.40 : 1 \| SHEET 4 OF 5 |

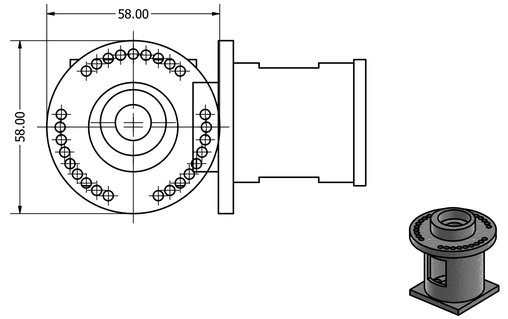
RA-115 J2BEARINGBLOCK
PA6-CF, QTY 1, 10 g (INVENTOR)
SCALE 1:1




RA-116 J2STANDOFF
PA6-CF, QTY 1, 26 g (INVENTOR)
SCALE 1:1



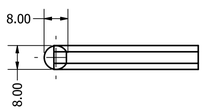
RA-117 J3DRIVEMOUNT
PA6-CF, QTY 1, 33 g (INVENTOR)
SCALE 1:1



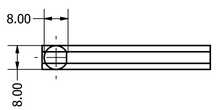

RA-118 J3ECCRING
PA6-CF, QTY 1, 3 g (INVENTOR)
SCALE 1:1




RA-212 J2JACKSHAFT
Steel, QTY 1, 18 g (INVENTOR)
SCALE 1:1




RA-213 J3JACKSHAFT
Steel, QTY 1, 21 g (INVENTOR)
SCALE 1:1




NOTES:
1. MATERIAL: PA6-CF (BAMBU LAB) UNLESS NOTED; PETG WHERE NOTED. DRY PA6-CF 80 C / 8 H.
2. PRINT: 0.4 MM NOZZLE, 0.2 MM LAYERS, 5 WALLS, 40% GYROID, 6 TOP/BOTTOM. ANNEAL PA6-CF 80 C / 6 H.
3. BEARING SEATS PRINTED -0.10 MM, PRESS FIT AFTER TEST PRINT. M3/M4 BRASS HEAT-SET INSERTS WHERE 4.0 / 5.6 HOLES.
4. GEARS: 20 DEG PRESSURE ANGLE, PRINT FLAT (TEETH IN XY). UNTOLERANCED DIMS +/-0.2 MM.
5. ALL DIMENSIONS MM. SOURCE: _build/design.py + build_parts.py (PARAMETRIC SCRIPT).

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| | | SIZE D · DWG NO RA-DWG-100 · REV |
| | | SCALE 1 : 1 · SHEET 5 OF 5 |